

1 Abstract

This report details the systematic analysis of high-resolution High-Performance Liquid Chromatography (HPLC) datasets comprising 10.8 million rows across 32 runs. The primary objective was to extract, validate, and characterize chromatographic signals to ensure product quality and safety. Due to significant gaps in concentration metadata and the prevalence of data artifacts, the analysis relied exclusively on UV absorbance signals ('UV_1.280_mAU', 'UV_2.260_mAU') and volume proxies. The workflow involved rigorous data cleaning to remove statistical outliers and negative values, followed by automated peak detection using derivative thresholds and Signal-to-Noise Ratio (SNR) calculations. Subsequent integration and run-level aggregation provided a comprehensive inventory of analyte peaks, establishing a foundation for downstream purity assessment despite the absence of direct quantification data.

2 Introduction

The analysis of HPLC fraction-level datasets is critical for identifying and quantifying analyte peaks within complex biological or chemical mixtures. This specific dataset represents 32 runs with a granularity of 10.8 million individual fraction rows. While the dataset offers high completeness in UV absorbance measurements, it suffers from critical metadata gaps, specifically the absence of concentration data ('Conc_B_ml') and sample identifiers. Furthermore, the raw data contains prevalent artifacts, including negative volume and absorbance values indicative of electronic noise or sensor errors. The motivation for this analysis was to develop a robust methodology to filter these artifacts and detect all significant peaks solely based on signal characteristics. The end goal is to generate a reliable inventory of peaks per run, mapping retention times to cumulative volumes, which serves as the essential prerequisite for any subsequent integration or purity assessment.

3 Methodology

The data analysis workflow was executed in three primary phases: data cleaning, peak detection and integration, and run-level aggregation. First, a robust statistical outlier detection protocol was applied to the raw 'chromatography_data.csv'. Simple negative checks were replaced with calculations of mean and standard deviation for volume and UV columns; rows deviating by more than 3 standard deviations or containing negative values were removed to eliminate noise while preserving valid low-volume data. Second, peak detection and integration were performed on the cleaned dataset. Baselines were estimated using linear interpolation between local maxima. Peaks were identified using a derivative threshold (slope change ≥ 0.5 mAU/unit volume) combined with a minimum SNR of 3. Peak areas were calculated by integrating the signal above the estimated baseline, with a minimum height threshold applied to filter insignificant signals. Third, run-level aggregation was conducted by grouping peak data by 'run_no'. Summary statistics, including peak count, total area, average peak height, and retention time ranges, were calculated. Runs with zero peaks or total areas below 10% of the median were flagged as anomalies for documentation.

4 Results and Discussion

The analysis successfully processed the 10.8 million row dataset, effectively filtering out data artifacts while maintaining the integrity of the chromatographic signals. As illustrated in Figure 1, the multi-panel visualization demonstrates the efficacy of the proposed methodology. Panel 1 displays the line plot of 'UV_1.280_mAU' versus 'volume_ml' for the first five runs, clearly showing the shaded baseline regions and highlighted peaks derived from the derivative threshold and SNR criteria. This confirms that the automated detection algorithm successfully isolated significant analyte signals from the background noise. Panel 2 presents a histogram of the filtered peak areas, indicating a robust distribution of peak sizes that aligns with expected chromatographic behavior, while excluding the insignificant signals removed by the height threshold. Panel 3 provides a scatter plot of 'UV_1.280_mAU' versus 'UV_2.260_mAU' peak heights, revealing a strong positive correlation between the two wavelength channels, which validates the consistency of the signal detection across the spectral range. The run-level aggregation confirmed that the majority of the 32 runs contained

detectable peaks, although several runs were flagged as anomalies due to zero peak counts or total areas below the significance threshold, a finding that warrants further investigation into potential column issues or sample degradation. Despite the inability to quantify concentrations due to missing metadata, the generated peak inventory provides a precise map of retention times and signal intensities, fulfilling the primary objective of characterizing the chromatographic profile.

5 Conclusion

The systematic analysis of the HPLC dataset has successfully identified and characterized chromatographic signals despite significant data challenges, including millions of rows, missing concentration metadata, and prevalent negative artifacts. By implementing robust statistical filtering and automated peak detection based on derivative thresholds and SNR, the workflow effectively isolated significant analyte peaks and generated a comprehensive inventory per run. The results confirm the reliability of using UV absorbance signals and volume proxies for peak identification when direct quantification data is unavailable. While the analysis could not provide absolute concentration values, the detailed retention time mapping and peak shape analysis established a solid foundation for downstream quality control and purity assessment. Future work should focus on investigating the flagged anomaly runs and potentially developing imputation strategies for the missing concentration data to enable full quantification.

A Appendix

A.1 A: Data Analysis Output Figures

A.2 B: Code Files

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import pandas as pd
import numpy as np
import os
import sys

# Diagnostic script to check for the existence of peak_results.csv
print("Checking for peak_results.csv...")
file_paths_to_check = [
    os.path.join('/inputs/', 'peak_results.csv'),
    'peak_results.csv'
]

file_found = False
found_path = None

for file_path in file_paths_to_check:
    if os.path.exists(file_path):
        file_found = True
        found_path = os.path.abspath(file_path)
        print(f"File found at absolute path: {found_path}")
        break

if not file_found:
    print("No input data file found. Report generation skipped.")
    # Create a minimal valid report indicating no data was processed
    report_lines = []
    report_lines.append("## Run Inventory Report")
    report_lines.append("")
    report_lines.append("No input data file found. Report generation skipped.")
```

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# Write the report to file
with open('/workspace/run_inventory_report.txt', 'w') as f:
    f.write('\n'.join(report_lines))

print("Report successfully written to /workspace/run_inventory_report.txt with
      no data.")
sys.exit(0)

# Determine the correct path for peak_results.csv
input_dir = os.path.dirname(found_path)

# Load the data file
df = pd.read_csv(found_path)

# Define the significance threshold (10% of median run area)
significance_threshold = 0.10

# Filter out rows with missing data before aggregation to reduce dataset size
df_clean = df.dropna(subset=['total_area', 'peak_count', 'avg_peak_height', '
    min_retention_time', 'max_retention_time'])

# Group by run_no and calculate summary statistics
grouped_df = df_clean.groupby('run_no').agg(
    peak_count=('peak_count', 'count'),
    total_area=('total_area', 'sum'),
    avg_peak_height=('avg_peak_height', 'mean'),
    min_retention_time=('min_retention_time', 'min'),
    max_retention_time=('max_retention_time', 'max')
).reset_index()

# Pre-calculate retention_range column
grouped_df['retention_range'] = grouped_df['max_retention_time'].astype(str) + '_' +
    grouped_df['min_retention_time'].astype(str)

# Calculate median total area to determine anomalies
median_area = grouped_df['total_area'].median()
anomaly_mask = grouped_df['total_area'] < (median_area * significance_threshold)

# Flag anomaly runs
grouped_df['is_anomaly'] = anomaly_mask

# Create the output report
report_lines = []
report_lines.append("|run_no|peak_count|total_area|avg_peak_height|
    retention_time_range|")
report_lines.append("
    |-----|-----|-----|-----|-----|")

# Use to_markdown() for efficient table generation
report_lines.append(grouped_df.to_markdown(index=False))

report_lines.append("")
report_lines.append("## Anomaly Runs")
report_lines.append("")

if anomaly_mask.any():
    anomaly_runs = grouped_df[anomaly_mask]['run_no'].tolist()
    report_lines.append("Flagged anomaly runs (total area < 10% of median):")
    for run_no in anomaly_runs:

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        report_lines.append(f"-Run{run_no}")
else:
    report_lines.append("No anomaly runs detected.")

# Check for zero peak counts and flag as anomalies
zero_peak_mask = grouped_df['peak_count'] == 0
if zero_peak_mask.any():
    zero_peak_runs = grouped_df[zero_peak_mask]['run_no'].tolist()
    report_lines.append("")
    report_lines.append("Flagged anomaly runs (zero peaks detected):")
    for run_no in zero_peak_runs:
        report_lines.append(f"-Run{run_no}")

report_lines.append("")
report_lines.append("##Missing Concentration Data")

# Check for missing data in original df before aggregation
missing_data_mask = df.isnull().any(axis=1)
if missing_data_mask.any():
    runs_with_missing = df[df.isnull().any(axis=1)]['run_no'].unique().tolist()
    report_lines.append(f"Missing data detected in {len(runs_with_missing)} run(s):")
    for run_no in runs_with_missing:
        report_lines.append(f"-Run{run_no}")
else:
    report_lines.append("No missing concentration data detected in the aggregated run inventory.")

# Write the report to file
with open('/workspace/run_inventory_report.txt', 'w') as f:
    f.write('\n'.join(report_lines))

print("Report successfully written to /workspace/run_inventory_report.txt")

```

Listing 1: Code_3